OECD Economics Department Working Papers

Mapping drought severity in Mexico using high-resolution satellite data

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No. 1862

OECD

OECD Economics Department Working Papers

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ABSTRACT/RÉSUMÉ/RESUMEN

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This paper analyses drought severity across Mexican regions between 2000 and 2025 using satellite-based indicators of vegetation health and surface moisture. Unlike traditional station-based measures, which depend on sparse weather data and often involve reporting delays, this approach directly captures how ecosystems respond to water stress across the entire territory. By combining these signals into a regional Normalised Difference Drought Index, the study provides a consistent, high-resolution measure of drought intensity for all Mexican states, identifying both the location and severity of water stress. The methodology enables detection of drought conditions at a finer spatial scale than conventional indicators, revealing substantial intra-state variation, such as the contrasts observed within Querétaro. The results show a persistent north–south divide: northern states, characterised by arid and semi-arid climates, experience more frequent and severe droughts, while southern regions retain higher levels of vegetation and moisture. From an economic and fiscal perspective, these patterns point to greater vulnerability in the north, where water-intensive agricultural, industrial, and energy sectors are concentrated. The findings underscore the value of near-real-time, spatially detailed monitoring to support Mexico's climate adaptation policies and to inform the development of fiscal resilience mechanisms to mitigate climate-related risks.

Keywords: Drought monitoring, Remote sensing, Normalized difference vegetation index (NDVI), Normalized difference water index (NDWI), Normalized difference drought index (NDDI), Mexico

JEL codes: Q54, O13, R11, C55

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Cartographie de la gravité de la sécheresse au Mexique à l'aide de données satellitaires à haute résolution

Cet article analyse la sévérité des sécheresses dans les régions mexicaines entre 2000 et 2025 à l'aide d'indicateurs satellitaires de l'état de la végétation et de l'humidité de surface. Contrairement aux mesures traditionnelles fondées sur des stations météorologiques, qui reposent sur des données limitées et souvent publiées avec retard, cette approche capte directement la réponse des écosystèmes au stress hydrique sur l'ensemble du territoire. En combinant ces deux signaux au sein d'un indice régional normalisé de sécheresse (Normalised Difference Drought Index), l'étude fournit une mesure cohérente et à haute résolution de l'intensité de la sécheresse pour tous les États mexicains, permettant d'identifier à la fois sa localisation et sa gravité. Cette méthodologie permet de détecter les conditions de sécheresse à une échelle spatiale plus fine que les indicateurs conventionnels, mettant en évidence d'importantes variations intra-étatiques, comme celles observées au Querétaro. Les résultats révèlent une fracture persistante entre le nord et le sud : les États du nord, caractérisés par des climats arides et semi-arides, connaissent des sécheresses plus fréquentes et plus intenses, tandis que les régions du sud maintiennent des niveaux plus élevés de végétation et d'humidité. Du point de vue économique et budgétaire, ces dynamiques indiquent une vulnérabilité accrue dans le nord, où les secteurs agricole, industriel et énergétique sont particulièrement intensifs en eau. Les résultats soulignent l'intérêt d'un suivi quasi en temps réel et spatialement détaillé pour appuyer les politiques d'adaptation au changement climatique au Mexique et orienter la conception de mécanismes de résilience budgétaire face aux risques climatiques.

Mots-clés : Veille de la sécheresse, Télédétection, Indice de végétation par différence normalisée (NDVI), Indice d'eau par différence normalisée (NDWI), Indice de sécheresse par différence normalisée (NDDI), Mexique.

Codes JEL: Q54, O13, R11, C55

Mapear la severidad de la sequía en México utilizando datos satelitales de alta resolución

Este artículo analiza la severidad de la sequía en las regiones mexicanas entre 2000 y 2025 utilizando indicadores satelitales del estado de la vegetación y la humedad superficial. A diferencia de las medidas tradicionales basadas en estaciones meteorológicas, que dependen de datos limitados y suelen presentar retrasos en su publicación, este enfoque capta directamente la respuesta de los ecosistemas al estrés hídrico en todo el territorio. Al combinar estas señales en un Índice de Sequía de Diferencia Normalizada regional (Normalised Difference Drought Index), el estudio proporciona una medida consistente y de alta resolución de la intensidad de la sequía para todos los estados mexicanos, identificando tanto su localización como su gravedad.Esta metodología permite detectar condiciones de sequía a una escala espacial más fina que los indicadores convencionales, revelando importantes variaciones dentro de los estados, como las observadas en Querétaro. Los resultados muestran una persistente brecha norte-sur: los estados del norte, caracterizados por climas áridos y semiáridos, experimentan sequías más frecuentes e intensas, mientras que las regiones del sur mantienen mayores niveles de vegetación y humedad.Desde una perspectiva económica y fiscal, estos patrones indican una mayor vulnerabilidad en el norte, donde los sectores agrícola, industrial y energético son particularmente intensivos en el uso del agua. Los hallazgos subrayan el valor del monitoreo casi en tiempo real y con alto detalle espacial para fortalecer las políticas de adaptación al cambio climático en México y orientar el diseño de mecanismos de resiliencia fiscal frente a los riesgos climáticos.

Palabras clave : Monitoreo de sequía, teledetección, Índice de Vegetación de Diferencia Normalizada (NDVI), Índice de Agua de Diferencia Normalizada (NDWI), Índice de Sequía de Diferencia Normalizada (NDDI), México.

Códigos JEL: Q54, O13, R11, C55

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Mapping drought severity in Mexico using high-resolution satellite data

Ilyes Boumahdi and Alberto González Pandiella[1]

Introduction

Droughts impose rising economic costs worldwide through multiple interconnected channels. At the production level, water scarcity reduces agricultural yields, livestock productivity, and hydropower generation, while constraining industrial and service activities that rely on water-intensive processes. These supply shocks raise food and energy prices, generating inflationary pressures and reducing real household income. On the fiscal side, governments face lower tax revenues and higher spending demands for emergency relief, reconstruction, and subsidies, which can widen fiscal deficits and increase public debt.

The global economic cost of drought has increased at an average annual rate of at least 3%. Since 2000, these costs have more than doubled and are projected to rise by a further 40% by 2035 (OECD, 2025[1]). A moderate to severe drought episode reduces global GDP growth by 0.98 percentage points and government revenue by 0.47 percent of GDP per year. The impacts are even more pronounced in emerging and developing economies, where GDP falls by 1.42 percentage points and fiscal revenues by 0.66 percent of GDP, respectively (Fuje et al., 2023[2]). Moreover, disasters often trigger fiscal responses, as governments provide financial assistance or reconstruction support (OECD and World Bank, 2019[3]). These unplanned expenditures, coupled with lower revenues, can lead to significant increases in public debt (Fuje et al., 2023[2]).

Mexico is highly exposed to droughts, which severely affect water availability across key sectors. Agriculture accounts for 53% of total national water demand and electric generation for 26% (Conagua, 2020[4]). Regional exposure to prolonged droughts can exacerbate these pressures, especially in states dependent on water-intensive activities. Average national rainfall has already decreased by 14 millimeters (mm) in 2024 compared to the 1981-2010 baseline. Deficits are particularly high in the regions of Sinaloa (-323 mm) and Durango (-196 mm) (OECD, 2025[5]; OECD, 2025[6]). Currently 22% of Mexican hydrological basins and 73% of aquifers are in water deficit (Conagua, 2020[4]), underscoring the urgency of improving drought monitoring and management capacities.

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Most existing studies on drought in Mexico rely on data from meteorological stations, like precipitation and temperature, to derive drought indices (Romero et al., 2020[7]; Ortiz-Gómez et al., 2018[8]; Carrillo-Carrillo et al., 2025[9]; Bautista-Capetillo et al., 2016[10]; Velázquez-Zapata and Dávila-Ortiz, 2025[11]). However, such station-based analyses often suffer from limited spatial coverage and delayed reporting. Recent advances in remote sensing offer a complementary and scalable approach to assess drought severity using satellite-derived indicators of vegetation and surface moisture.

Similar methods have been applied in other regions, including the Flint Hills ecoregion and the Tallgrass Prairie National Preserve site, spanning southern Kansas to northern Texas in the United States (Gu et al., 2007[12]), as well as the Parbhani district in India (Patil et al., 2024[13]). To our knowledge, in Mexico, this approach has only been applied at a subregional scale, specifically in the Guadalupe Valley basin in Baja California (Del-Toro-Guerrero et al., 2022[14]) and a subregion of Veracruz state (Salas-Martínez et al., 2023[15]). By contrast, this paper covers the entire national territory, encompassing all Mexican states and municipalities, and provides retrospective analysis spanning the period from 2000 to 2024 and prospective analysis for 2025.

Building on this literature, the paper adopts a remote-based approach to measure drought severity. We extract and process satellite imagery to construct a regional drought index (Sections 2 to 4). Using updated data at 8-day resolution, we then present an illustrative nowcast of drought severity across Mexican states for 2025, intended to demonstrate how these techniques can be used to provide timely approximations of drought conditions before complete annual data become available (Section 5). The paper concludes with a discussion of the main findings in Section 6.

Data and methods

This study employs a remote sensing–based approach to monitor drought severity across all Mexican states. Remote sensing refers to the acquisition of information about the Earth's surface without direct contact. In this study, it is operationalised through satellite imagery obtained from multiple sensors (e.g. MODIS, Landsat, Sentinel). These data sources enable consistent, spatially comprehensive, and high-frequency monitoring of vegetation and surface conditions, thereby overcoming key limitations of meteorological station networks, such as uneven spatial coverage, limited temporal continuity, and delays in data availability. We extract, through the Google Engine Earth (GEE), the Terra MODIS 8-day composite dataset (MOD09A1, Collection 4), which provides global coverage at 500-meter spatial resolution across seven spectral bands (bands-1-7) (USGS, 2025[16])). Three of these bands were used to derive the indicators required for the analysis, namely the near-infrared band (B_{NIR}, wavelength 841 to 876 nanometer (nm)), the red band (B_{Red}, wavelength 620 to 670 nm) and the short-wavelength infrared band (B_{SWIR}, wavelength 2105 to 2155 nm).

The Moderate Resolution Imaging Spectroradiometer (MODIS) was selected for its long temporal record (available from 2000 to 2025, with a data latency of no more than four weeks at the time of extraction), high revisit frequency, and robust atmospheric corrections. Each 8-day composite represents the best-quality surface reflectance observation within that period, after filtering for clouds, aerosols, and sensor noise.

The use of GEE enables automated batch processing and spatial aggregation of large image collections across all Mexican states, ensuring methodological consistency over time and space. This information was used to Normalised Difference Vegetation Index (NDVI), the Normalised Difference Water Index (NDWI), and the Normalised Difference Drought Index (NDDI), which serve as proxies for vegetation health, surface moisture, and overall drought severity, respectively.

Healthy vegetation absorbs most visible red radiation for photosynthesis while strongly reflecting near-infrared radiation. By contrast, stressed or sparse vegetation reflects less near-infrared energy. This spectral contrast provides a reliable basis for identifying areas affected by drought (NASA, 2025[17]). The normalized difference vegetation index (NDVI) captures this contrast by comparing reflected near-infrared radiation with absorbed red radiation (Eq 1) NDVI has values ranging between -1 and 1:

$$\mathrm{NDVI} = \frac{\mathrm{B_{NIR}} - \mathrm{B_{Red}}}{\mathrm{B_{NIR}} + \mathrm{B_{Red}}} \qquad \text{Eq. 1}$$

Where:

B_{NIR}: Near-infrared band (841-876 nm wavelength).

B_{Red}: Red band (620-670 nm).

Furthermore, so wetter areas look darker in satellite images. This spectral contrast enables the identification of surface and vegetation moisture conditions (NASA, 2025[17]). Building on this principle, we compute the normalized difference water index (NDWI) as a difference between reflected and absorbed radiation in the near-infrared and short-wave infrared bands(Eq. 2). NDWI is a widely used indicator of vegetation water content and surface moisture and ranges from −1 to 1:

$$\mathrm{NDWI} = \frac{\mathrm{B_{NIR}} - \mathrm{B_{SWIR}}}{\mathrm{B_{NIR}} + \mathrm{B_{SWIR}}} \qquad \text{Eq. 2}$$

Where:

B_{SWIR}: Short-wavelength infrared band (2105-2155 nm wavelength).

NDVI and NDWI reflect the actual condition of vegetation and surface moisture, not rainfall alone. Irrigated cropland may therefore remain relatively green during meteorological drought because water inputs offset rainfall deficits. The indices thus capture observed vegetation stress, combining climatic conditions and human water management. NDVI has also been widely used to identify irrigated cropland and irrigation intensity (e.g. Zhang et al., 2022), supporting its relevance for analysing drought dynamics in managed landscapes.

In addition to spectral indices, we estimate evapotranspiration as a complementary measure of drought dynamics. Evapotranspiration captures the combined processes of water evaporation from soil and transpiration from vegetation and plays a key role in determining soil moisture availability and the resilience of relatively humid areas to drought conditions.

Evapotranspiration (Eq.3) is calculated as a function of annual precipitation P (mm/year) and mean annual temperature T (°C) (Cárdenas-Tristán et al., 2023[18]):

$$E = \frac{P}{\sqrt{0.9 + \frac{P^2}{\tau^2}}} \qquad \text{Eq. 3}$$

Where τ (Eq.4) is a parameter based on the annual average temperature T:

$$\tau = 300 + 25\mathrm{T} + 0.05\mathrm{T}^3 \qquad \text{Eq. 4}$$

Higher temperatures increase τ, indicating greater energy availability for evaporation and plant transpiration. This formulation allows the model to capture how climatic warmth influences potential water losses from soil and vegetation, thereby amplifying drought vulnerability.

Finally, we calculated the Normalized Difference Drought Index (NDDI) (Eq. 5) by combining the NDVI (Eq.1) with NDWI (Eq. 2):

$$\mathrm{NDDI} = \frac{\mathrm{NDVI} - \mathrm{NDWI}}{\mathrm{NDVI} + \mathrm{NDWI}} \qquad \text{Eq. 5}$$

The Normalised Difference Drought Index (NDDI) combines information from both vegetation and water indices to identify drought conditions more accurately. When vegetation health (NDVI) is low and surface moisture (NDWI) is also low, NDDI values increase, indicating drier conditions. In contrast, healthy and well-watered areas show lower NDDI values. This makes the index a useful tool for distinguishing between regions experiencing mild, moderate, or severe drought.

Drought severity is classified according to NDDI values, following the threshold ranges proposed by (Patil et al., 2024[13]) :

- No drought: less than 0.2
- Mild drought: 0.2 - 0.3,
- Moderate drought 0.3 - 0.4
- Severe drought 0.4 - 0.5
- Extreme drought: more than 0.5

The NDDI classification relies on absolute index values rather than deviations from a long-term climatological mean. The indicator therefore captures observed surface dryness and vegetation stress at a given point in time, rather than precipitation anomalies relative to historical averages. The use of established threshold ranges provides a consistent framework for identifying mild to extreme drought conditions across regions and years.

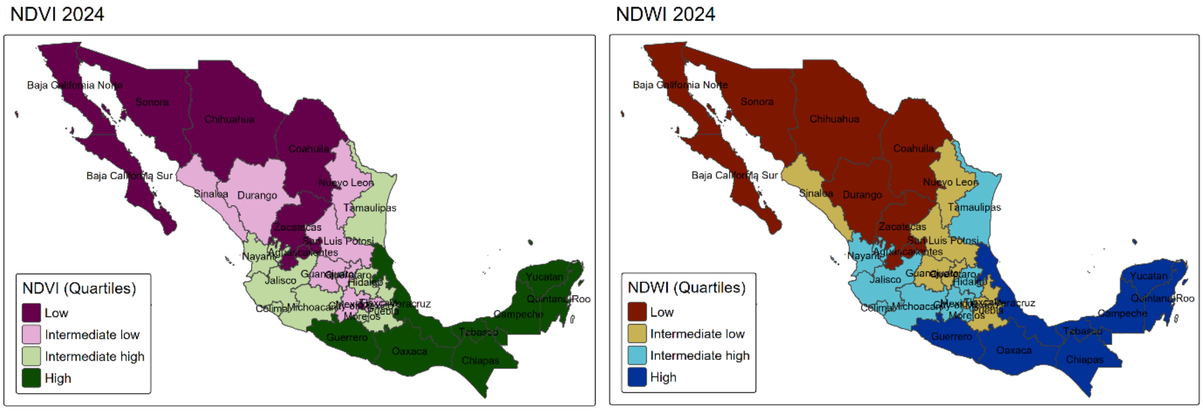
The data can be extracted and processed at a spatial resolution of 500 m², allowing for a high degree of geospatial disaggregation. However, to align the analysis with the scale at which regional public policies are designed, implemented, and monitored in Mexico, results are aggregated at two administrative levels: all Mexican states, corresponding to large regions under the TL2 geographic classification, and the municipalities of the state of Querétaro, corresponding to small regions under the TL3 classification.

Drought severity, exposure, and vulnerability across regions

Computing the indices described in the previous section reveals clear spatial and temporal patterns in drought severity across Mexico. The spatial distribution of vegetation and surface water conditions points to a pronounced north–south divide (Figure 1). Southern states such as Chiapas, Oaxaca, Veracruz, and Tabasco display high values of the Normalised Difference Vegetation Index (NDVI) and the Normalised Difference Water Index (NDWI), indicating relatively healthy vegetation and higher surface moisture. In contrast, northern states—including Baja California, Sonora, Chihuahua, and Coahuila—exhibit systematically lower NDVI and NDWI levels, reflecting drier climatic conditions and more sparsely vegetated landscapes. These two indicators are strongly correlated (Figure A1), suggesting that regions with more abundant vegetation also tend to maintain higher surface water availability.

Figure 1: Vegetation and water conditions in Mexico (2024)

Panel A. Normalised Difference Vegetation index   Panel B. Normalised Difference Water index

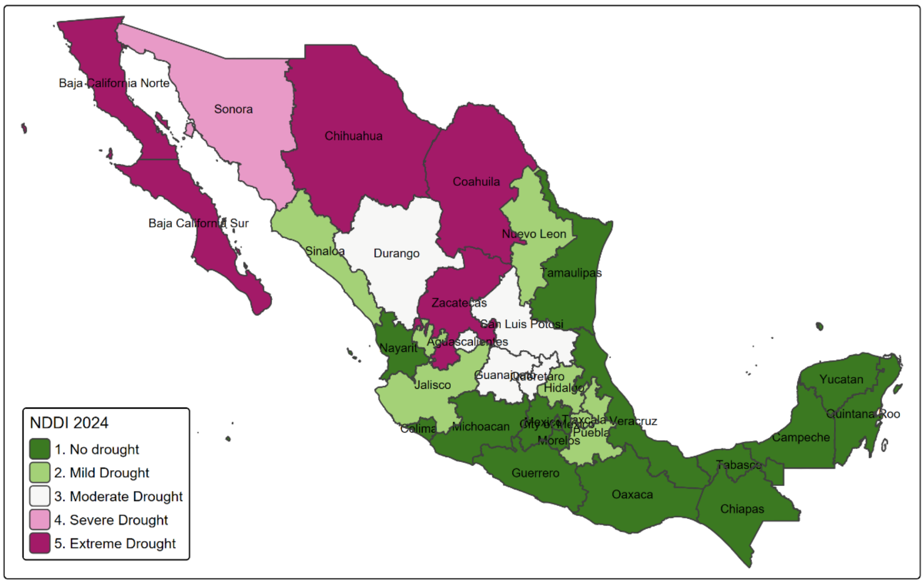


Source: Authors.

Drought severity, as measured by the Normalised Difference Drought Index (NDDI), follows a similar geographical pattern (Figure 2). Northern and north-central states, particularly Baja California Sur, Coahuila, Chihuahua, and Zacatecas, experience more severe drought conditions, whereas southern and south-eastern states generally face mild or no drought. These spatial patterns are consistent with earlier studies identifying northern and north-western Mexico as the country's most drought-prone regions, while the south and south-east benefit from higher rainfall and more favourable hydrological conditions (Méndez and Magaña, 2010[19]; Ortega-Gaucin, De la Cruz Bartolón and Castellano Bahena, 2018[20]).

Figure 2: Drought severity across Mexican states (2024)

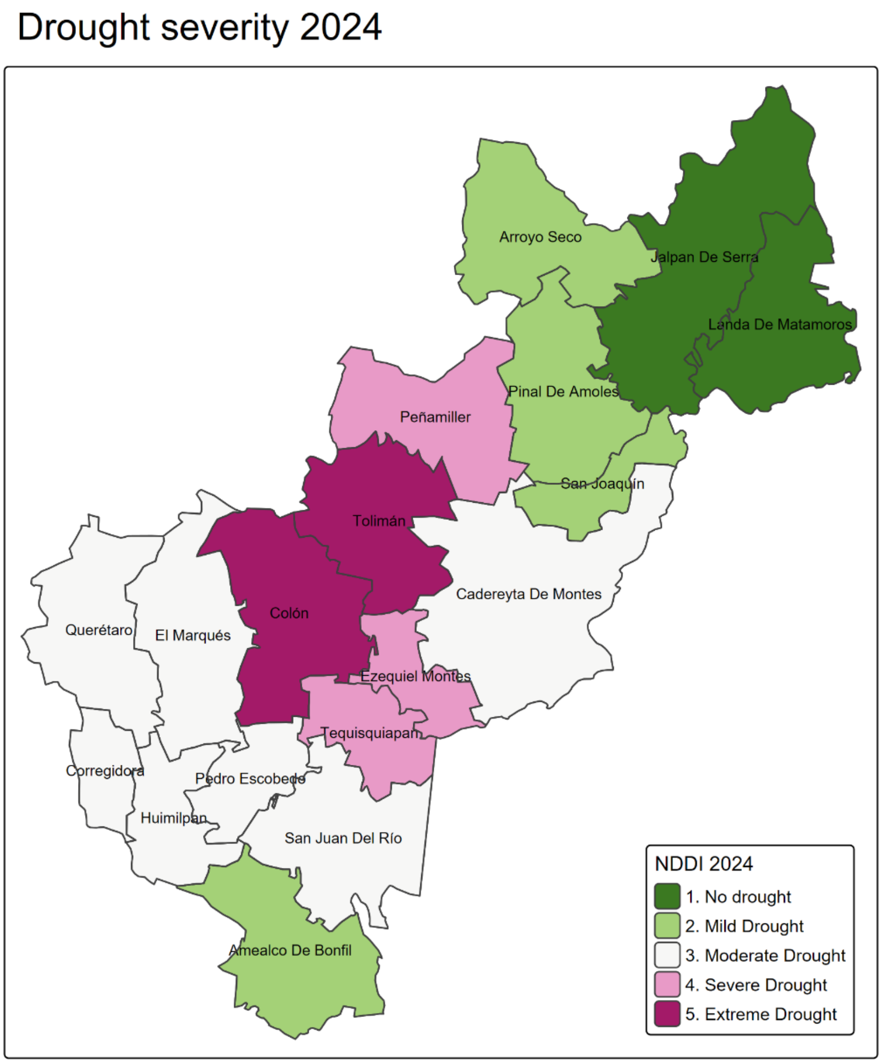
Drought severity 2024



Note: Drought severity classification is based on NDDI as follow: No drought (less than 0.2), mild drought]0.2,0.3], moderate drought]0.3,0.4], severe drought]0.4,0.5], extreme drought (more than 0.5) (Patil et al., 2024[13]).
Source: Authors.

One of the main advantages of satellite-based indicators is their high geographical granularity, which allows drought conditions to be assessed at very fine spatial scales and reveals heterogeneity that is often masked in conventional meteorological datasets. The case of Querétaro illustrates the importance of this level of detail (Figure 3). Despite being one of Mexico’s smallest states, Querétaro exhibits marked intra-state variation in drought severity across its 18 municipalities. In 2024, the central-western municipalities of Colón and Tolimán, located in the semi-arid *Semidesierto Queretano*, experienced extreme drought conditions. By contrast, the north-western municipalities of Jalpan de Serra and Landa de Matamoros, situated in the wetter and more temperate Sierra Gorda region, recorded no drought. This contrast underscores the value of remote sensing for identifying localised vulnerabilities and informing targeted water management and adaptation policies.

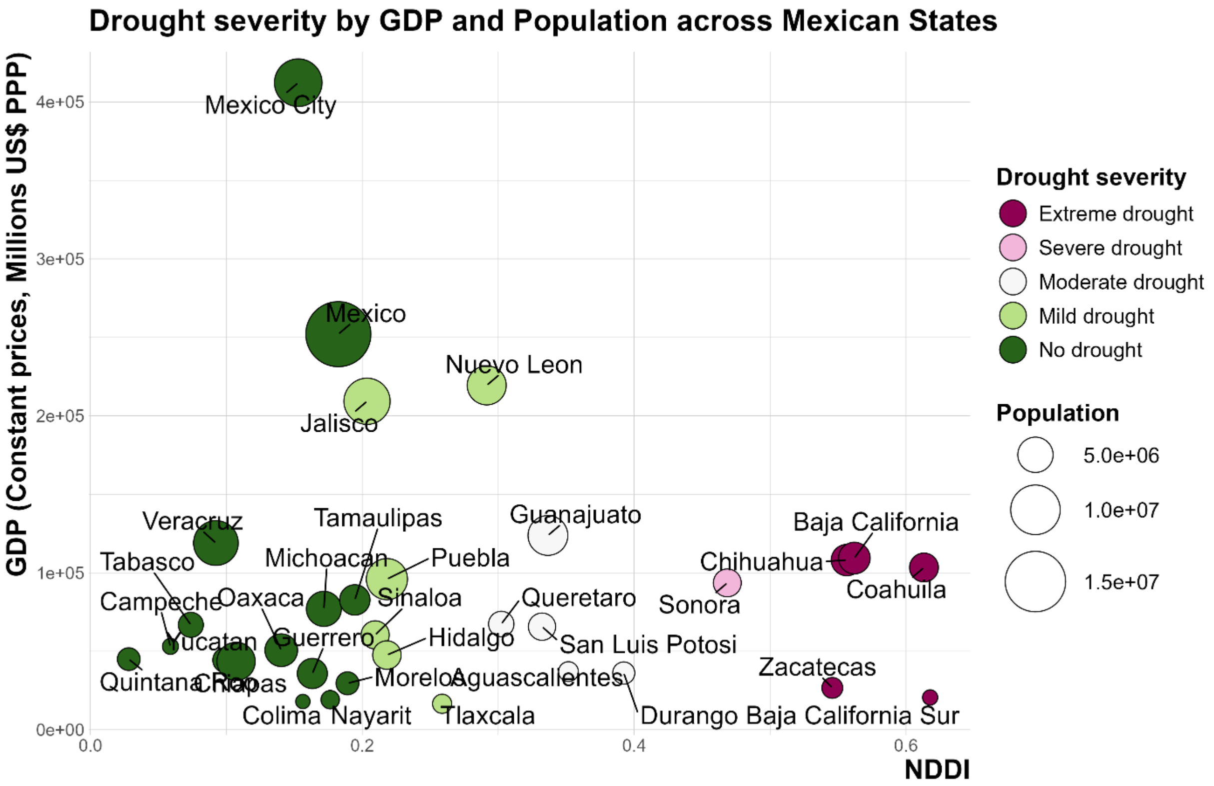
Figure 3: Drought severity across Queretaro municipalities (2024)



Source: Authors.

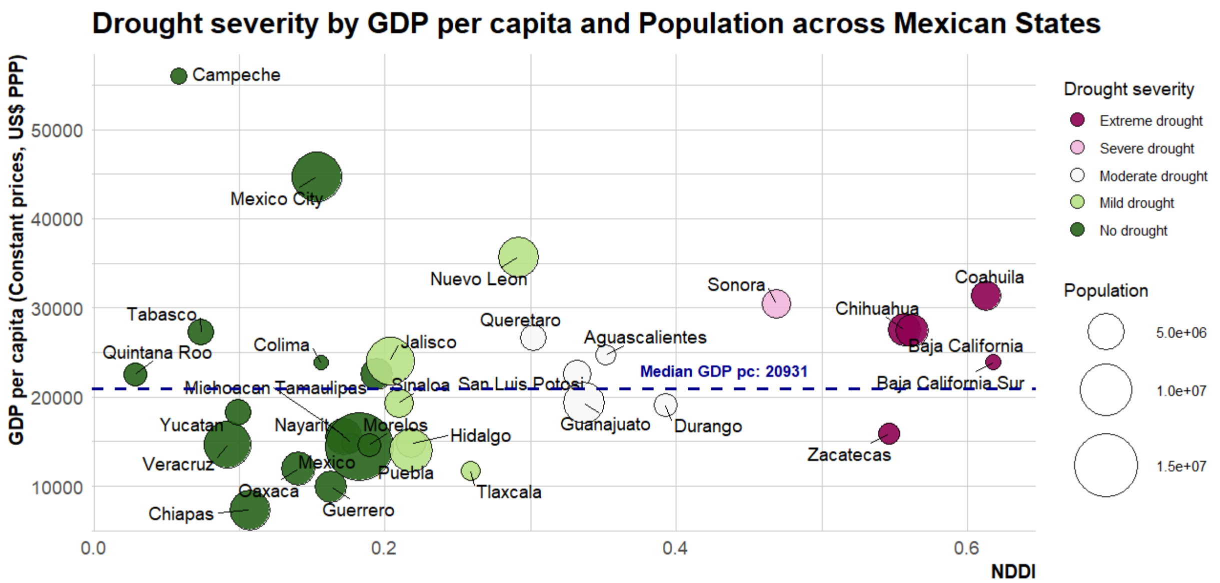
Although the states most exposed to drought account for a relatively modest share of national GDP and employment (Figures 4 and 5), this does not imply that drought risk is of limited economic or social importance. On the contrary, these regions tend to have lower GDP per capita and productivity levels (Figure 5 and 6), more limited fiscal capacity, and higher dependence on climate-sensitive activities such as agriculture and energy production. As a result, drought shocks can have disproportionate local impacts, exacerbating regional inequalities and placing sustained pressure on subnational public finances. Moreover, supply-chain linkages and internal migration mean that severe droughts in sparsely populated regions can still generate spillovers to the broader national economy.

Figure 4: Drought severity by GDP and population across Mexican States (2024)



Source: Authors.

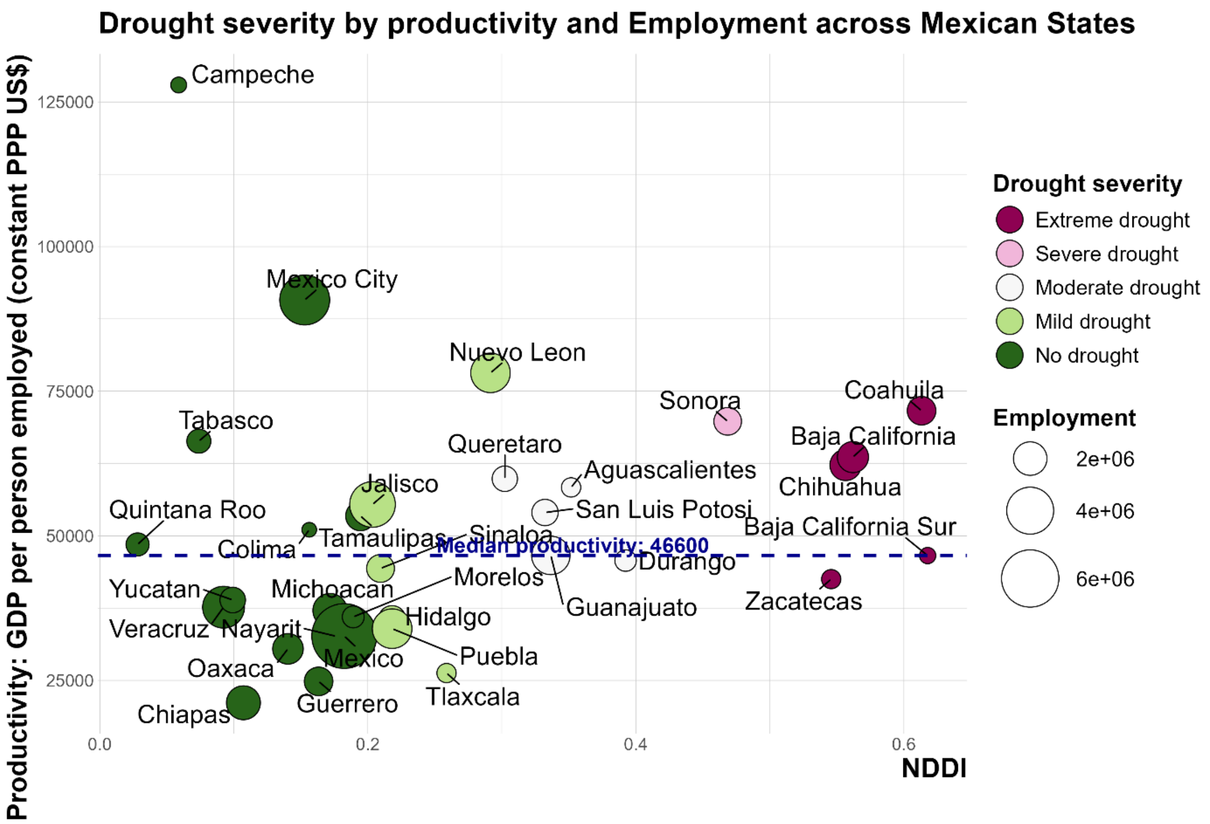
Figure 5: Drought severity by GDP per capita and population across Mexican States (2024)



Source: Authors.

Similarly, Zacatecas is the most vulnerable to drought in terms of productivity, as its GDP per person employed (42515 US$ PPP) is lower than the national median of (46600 US$ PPP). Furthermore, 11% of employment in Mexico are exposure to drought while states with no drought concentrate 53% of employment (Figure 6).

Figure 6: Drought severity by productivity and employment across Mexican States (2024)

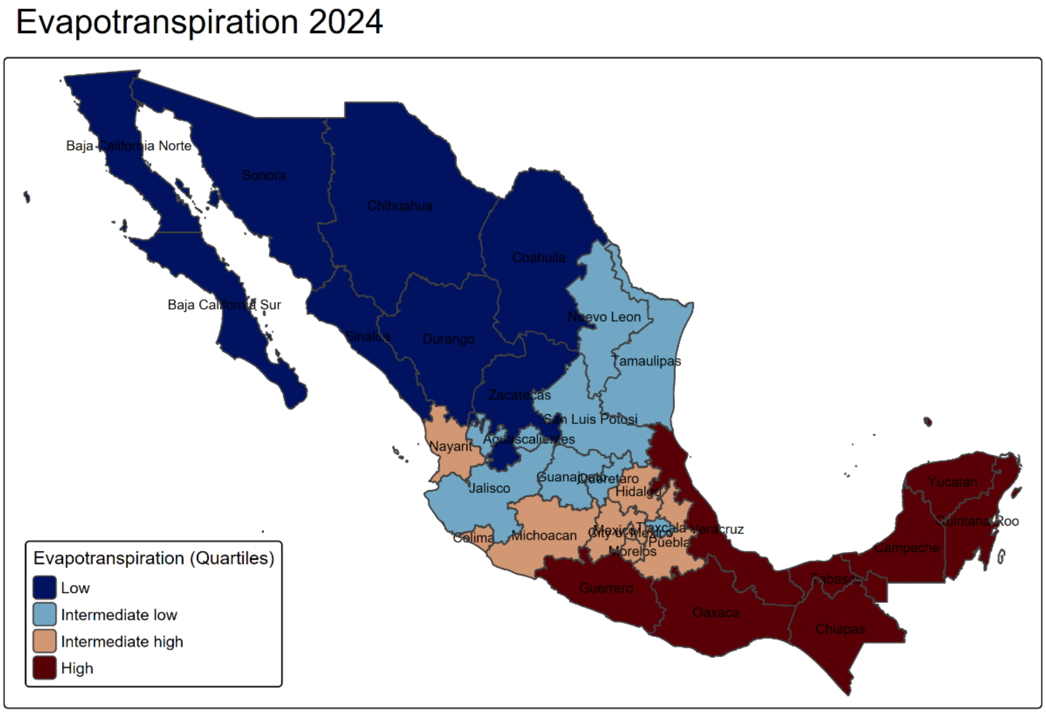


Source: Authors.

While southern states generally benefit from more favourable climatic conditions, they also exhibit higher evapotranspiration rates, (Figure 7), particularly in Guerrero, Oaxaca, Veracruz, and Chiapas. This reflects both greater water availability and higher temperatures. Elevated evapotranspiration indicates stronger water losses to the atmosphere through evaporation and plant transpiration. Although this currently coexists with relatively abundant rainfall, rising temperatures associated with climate change may increase potential water losses, amplifying vulnerability if precipitation becomes more variable or declines. Even regions that are presently less drought-prone may therefore face increasing water stress under sustained warming.

Taken together, these findings point to a dual drought profile in Mexico. The north-west exhibits structurally arid baseline conditions, which increase its exposure and sensitivity to drought episodes. Although drought intensity varies over time, as illustrated by the improvement observed in Sonora in 2025, shifting from extreme to severe drought, the region remains more vulnerable to severe water stress due to its underlying climatic profile. Other regions experience drought more episodically but may face increasing vulnerability due to climatic warming and rising evapotranspiration, underscoring the need for early-warning systems and locally tailored adaptation strategies.

Figure 7: Evapotranspiration levels across Mexican states (2024)



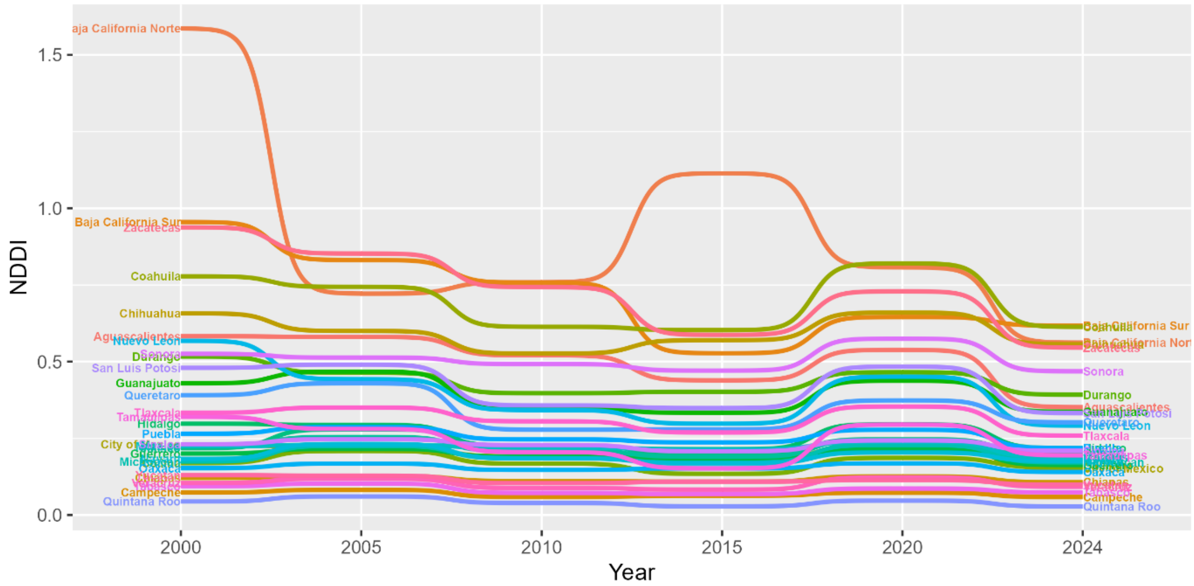
Source: Authors.

Evolution and persistence of drought severity over time

The temporal analysis of drought severity between 2000 and 2024 indicates a gradual reduction in average NDDI values in several regions, suggesting relative improvements in vegetation conditions and surface moisture compared to the early 2000s. This trend partly reflects the fact that the early 2000s were characterised by several exceptionally severe and spatially heterogeneous drought episodes across Mexico, which disproportionately raised baseline drought indicators during the initial years of the period. By contrast, subsequent years, including around 2010, the mid-2010s, and parts of the early 2020s, featured episodes of above-average rainfall in some regions, particularly in central and southern Mexico, contributing to temporary recoveries in vegetation and surface moisture conditions.

For example, Guerrero experienced a severe drought episode in 2015 that resulted in approximately USD 26 million in economic damages (OECD and World Bank, 2019[3]). Nonetheless, by 2024, drought conditions appeared less severe on average than in the early 2000s, with a mean reduction in NDDI of 16.7% relative to 2000. The largest declines were observed in Baja California Norte (–64.6%), Nuevo León (–41.8%), and Zacatecas (–41.8%) (Figure 8). However, these reductions should be interpreted cautiously, as several states, particularly in northern Mexico, continue to experience persistently high drought severity, reflecting enduring structural exposure rather than a sustained shift towards wetter conditions. The year 2024 marks the largest improvement in the last decade (with an average 16.7% reduction in NDDI relative to 2000). The most notable decreases occurred in Baja California Norte (-64.6%), Nuevo Leon (-41.8%) and Zacatecas (-41.8%) (Figure 8).

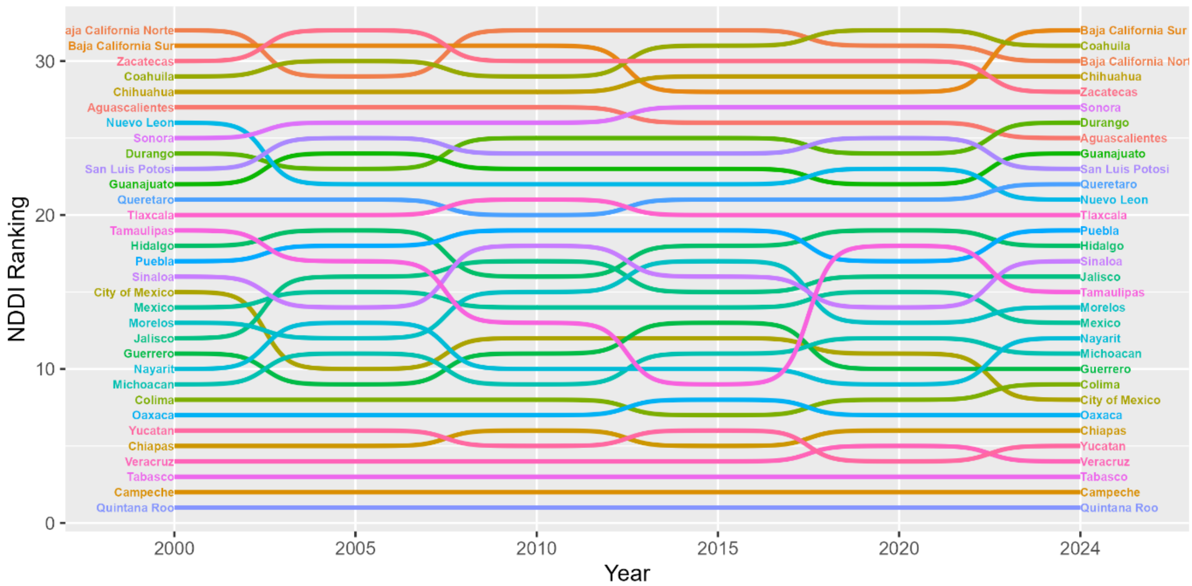
Figure 8: Evolution of drought severity by state between 2000 and 2024



Source: Authors.

Despite these changes in average drought intensity, the relative drought exposure and ecological stress patterns have remained largely stable over the past two decades (Figure 9). Northern states such as Baja California Sur, Coahuila, Baja California Norte, Chihuahua, and Zacatecas consistently exhibit the highest drought risk, while southeastern states including Quintana Roo, Tabasco, and Campeche remain among the least affected. This persistence underscores the structural nature of drought vulnerability in Mexico, shaped by enduring north–south asymmetries in climate, water availability, and ecosystem conditions.

Figure 9: Persistence of drought vulnerability drought across Mexican states between 2000 and 2024



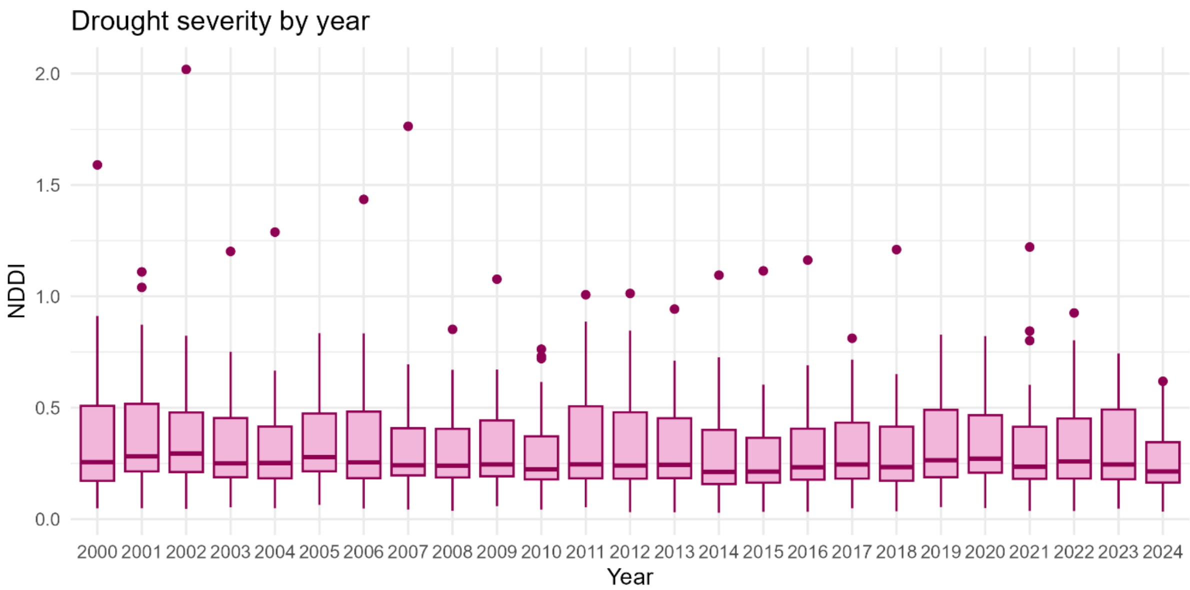
Source: Authors.

It is important to note that these indicators capture climatic exposure and observed ecological drought conditions, but they do not directly measure institutional resilience, water management capacity, or adaptive investments undertaken by individual states. Differences in infrastructure, groundwater reserves, irrigation efficiency, or fiscal buffers may mitigate or amplify the socio-economic impacts of similar drought intensities. The results should therefore be interpreted as reflecting relative exposure to drought conditions rather than the effectiveness of state-level resilience efforts.

The distribution of drought severity also varies markedly over time. During relatively less dry years—such as 2010, 2014, 2015, and 2024—the distribution of NDDI values becomes more concentrated, as reflected in a narrower interquartile range and a smaller number of extreme outliers (Figure 10). This pattern suggests that drought conditions during these years were more spatially homogeneous, with fewer states experiencing exceptionally severe stress relative to the national average. Such years are often associated with more favourable or evenly distributed climatic conditions, which tend to reduce disparities in vegetation and surface moisture across regions.

By contrast, the early 2000s—particularly 2000, 2002, 2006, and 2007—are characterised by much wider distributions and a larger number of high outliers, with NDDI values exceeding 1.5 and, in some cases, 2.0. These distributions indicate the coexistence of extreme drought conditions in some regions alongside more moderate conditions elsewhere, resulting in pronounced spatial heterogeneity. Periods combining high drought intensity with strong regional divergence are especially challenging from a policy perspective, as they tend to amplify agricultural losses, strain water supply systems, and generate uneven fiscal pressures across subnational governments. Distinguishing between years of broadly shared drought stress and those dominated by highly localised but severe episodes is therefore essential for effective drought risk management, early-warning systems, and the design of targeted adaptation policies.

Figure 10: Distribution of drought severity between 2000 and 2024

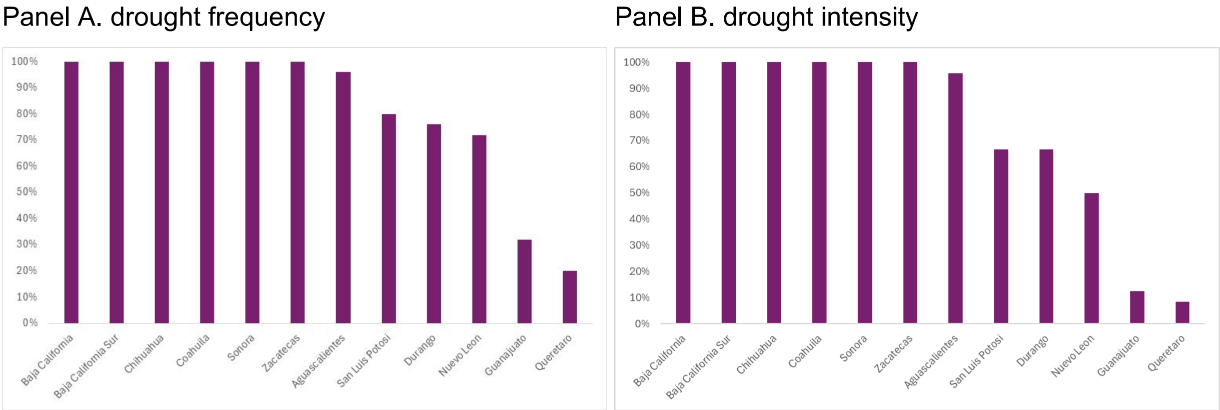


Source: Authors.

Over the full period 2000–2024, twelve Mexican states experienced at least one episode of severe to extreme drought. Six north-western states—Baja California, Baja California Sur, Chihuahua, Coahuila, Sonora, and Zacatecas—exhibit the highest drought frequency, with at least one municipality experiencing severe or extreme drought every year of the analysis (Figure 11, Panel A). Central and southern states show substantially lower frequencies. A similar spatial contrast emerges for drought persistence, defined as the share of years with at least two consecutive drought episodes (Figure 11, Panel B). Persistence is highest in the north-west, reflecting chronic aridity and limited water storage capacity. Persistent droughts are particularly concerning, as they erode water reserves, heighten agricultural vulnerability, and generate cumulative fiscal pressures, whereas more episodic droughts tend to produce irregular but less entrenched risks.

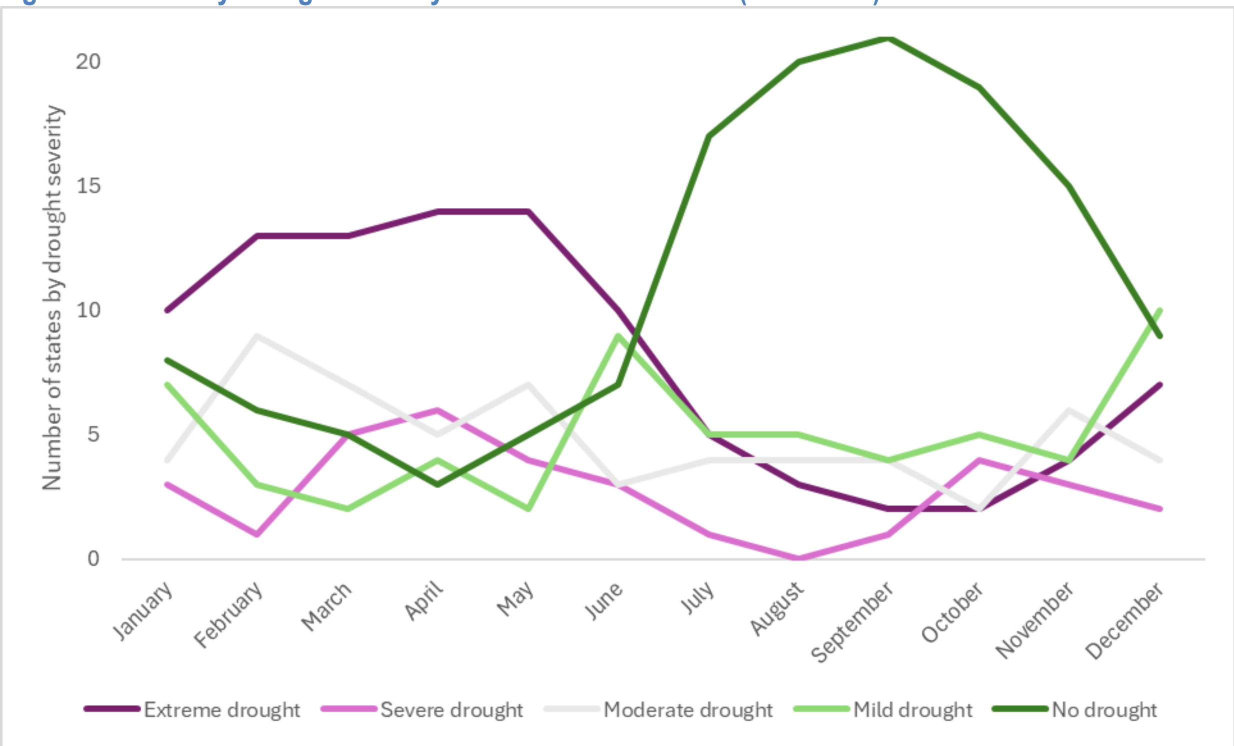
Drought severity also varies seasonally. During the August–September period (monthly mean, 2020–2024), only three states—Baja California, Coahuila, and Baja California Sur—experience severe to extreme drought conditions, compared with at least eighteen states during the March–May period (Figure 12). This seasonal pattern reflects the influence of precipitation cycles and highlights the importance of timing in drought monitoring and response.

Figure 11: Drought frequency and intensity in Mexico (2000-2024)



Source: Authors.

Figure 12: Monthly drought severity across Mexican states (2020-2024)



Note: Monthly mean value from 2020 to 2024.
Source: Authors.

Nowcasting drought conditions

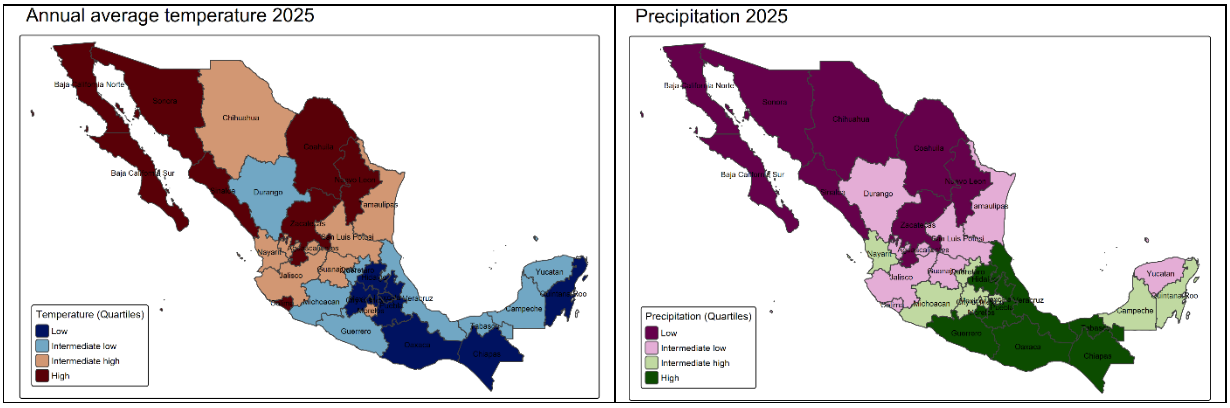
Accurate nowcasting of drought conditions, that is, producing near-real-time estimates based on the latest available satellite data, can be very valuable for economic stability and fiscal planning. Drought shocks can depress agricultural output, raise food and energy prices, and reduce tax revenues, while emergency relief and reconstruction spending put pressure on public budgets. Anticipating these dynamics enables governments to design early-warning systems, allocate contingency resources, and protect vulnerable sectors and households.

In this paper, the nowcast for 2025 is presented for illustrative purposes, with the aim of demonstrating how satellite-based indicators can be used to produce timely approximations of drought conditions for the remaining months of the year, before full hydrological-year data become available. The exercise does not seek to provide a definitive assessment of drought outcomes in 2025, but rather to highlight the potential of near-real-time monitoring to inform policy decisions under uncertainty.

To construct the 2025 nowcast, we use satellite-based observations of vegetation and surface moisture available up to August 2025. Specifically, MODIS 8-day composite data on surface reflectance, vegetation, and water content (Bands 1–7) are aggregated into monthly averages and used to compute the Normalised Difference Vegetation Index (NDVI), the Normalised Difference Water Index (NDWI), and, subsequently, the Normalised Difference Drought Index (NDDI), following the same methodology applied for previous years. Relying on data available through August allows for a timely approximation of annual drought conditions while preserving sufficient information to capture mid-year climatic dynamics.

The results suggest that Mexico's overall climatic configuration remains similar to that of 2024, with more favorable rainfall and moderate temperatures concentrated in southern states (Figure 13).

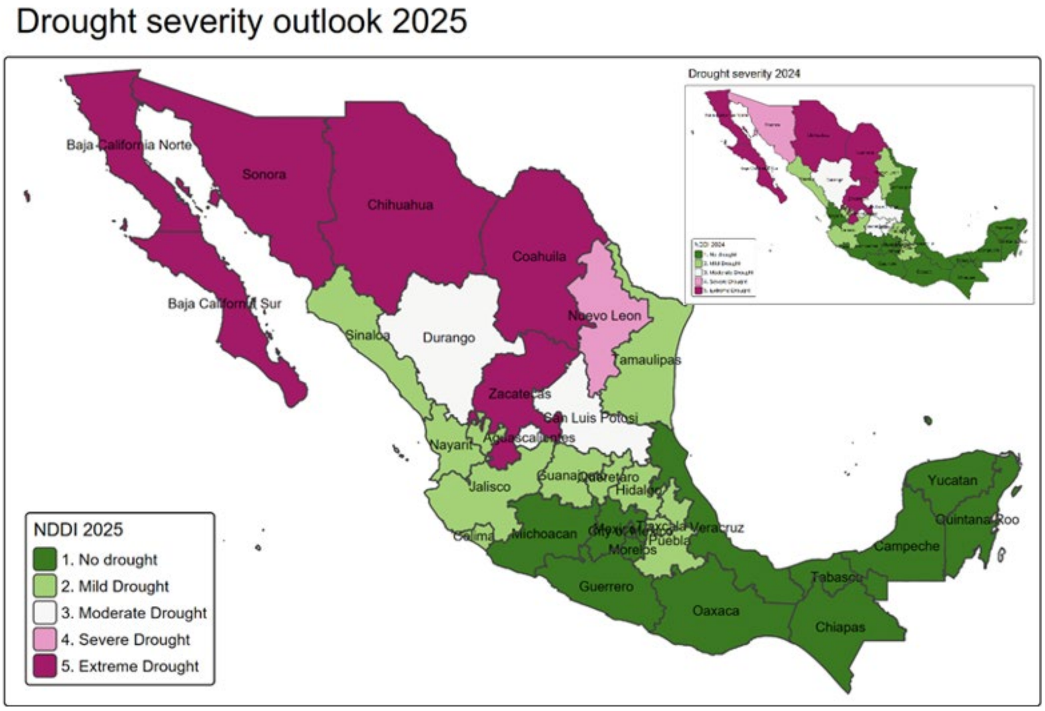
Figure 13: Climatic conditions in Mexico, 2025



Source: Authors.

Based on the Normalised Difference Drought Index (NDDI) computed from these partial-year data, the nowcast indicates limited changes in drought severity relative to 2024 for most states (Figure 14). But there are some interesting changes that illustrates the usefulness of the nowcasting exercise. Sonora shows a modest improvement, shifting from extreme to severe drought, while Guanajuato and Querétaro move from moderate to mild drought. Conversely, Nuevo León deteriorates from mild to severe drought, and Tamaulipas, Nayarit, and Colima shift from no drought to mild drought. For most other states, drought classifications remain unchanged, reinforcing the persistence of Mexico's north–south climatic divide.

Figure 14: Drought severity would not change for most of Mexican states in 2025



Note: Drought severity classification is based on NDDI as follow: No drought (less than 0.2), mild drought]0.2,0.3], moderate drought]0.3,0.4], severe drought]0.4,0.5], extreme drought (more than 0.5) (Patil et al., 2024[13]).
Source: Authors.

Monthly data for the most drought-exposed states broadly confirm this stability between 2024 and 2025 (Figure 15). Baja California Sur constitutes a partial exception: drought conditions were more severe during September–October 2024 than in the same months of 2025, while the period from January to April 2025 exhibits more pronounced drought intensity than in early 2024. Sub-monthly (8-day) data show a similar temporal profile, indicating that the nowcasted changes are consistent across temporal resolutions (Figure A2).

Figure 15: The monthly severity of drought does not change overall for the most exposed Mexican states in 2025

Source: Authors.

Conclusions

This paper provides an assessment of drought severity across Mexico from 2000 to 2025 using a remote sensing–based approach that combines satellite-derived vegetation and surface water indices. By leveraging consistent and high-frequency observations from the MODIS Terra platform, we construct a regional drought index (NDDI) that captures spatial and temporal variations in observed drought conditions with greater coverage than traditional meteorological station data. An important advantage of this approach is its high geographical granularity, which allows drought indicators to be computed at sub-regional and municipal levels, revealing local heterogeneity that may not be visible in conventional datasets. The pronounced intra-state contrasts observed in Querétaro, where neighbouring municipalities simultaneously experience extreme drought and no drought, illustrate the potential value of this fine spatial detail for targeted water management and adaptation planning. This methodological approach may contribute not only to understanding Mexico's recent drought patterns but also to inform climate change adaptation planning.

The results reveal a persistent north–south gradient in observed drought conditions, with the arid and semi-arid northern states, such as Baja California Sur, Coahuila, Chihuahua, and Zacatecas, facing recurrent and severe drought episodes. Southern and southeastern regions, including Tabasco, Campeche, and Quintana Roo, continue to benefit from higher rainfall, denser vegetation, and greater surface moisture. However, despite these favourable conditions, southern states also exhibit higher evapotranspiration levels, reflecting warmer and wetter conditions. While this currently coexists with relatively favourable hydrological conditions, rising temperatures associated with climate change could increase potential water losses and heighten sensitivity to future precipitation variability. Strengthening water storage capacity, together with more efficient water allocation and demand management, will be essential to prevent potential deficits under a warming climate.

The 2025 nowcast is presented as an illustrative exercise intended to demonstrate the potential of satellite-based monitoring for near-real-time drought assessment. The results suggest that Mexico's overall drought configuration in 2025 remains broadly consistent with that of 2024, but with some changes in most states. The exercise highlights how remote sensing can provide early signals of changes in drought severity, supporting anticipatory policy responses.

From a policy standpoint, the findings underscore the potential usefulness of integrating near-real-time drought monitoring within Mexico's climate adaptation and fiscal frameworks. Remote sensing offers a cost-effective and scalable tool for tracking drought evolution, supporting early-warning systems, and guiding targeted adaptation interventions. Strengthening institutional coordination between climate, water, and fiscal authorities will be crucial to ensure that budgetary responses are timely, data-driven, and adapted to regional needs.

The material presented in this paper is particularly relevant in the context of climate change, which is expected to intensify drought risks in Mexico. National projections indicate that average temperatures will rise by 0.34°C between 2030 and 2060 relative to 1981–2010, with increases ranging from 0.3°C in Aguascalientes, Yucatán, and Zacatecas to 0.5°C in Michoacán (OECD, 2025[6]; OECD, 2025[5]). Rising temperatures are likely to accelerate evapotranspiration and further constrain water availability, increasing the value of tools capable of monitoring vegetation stress and surface moisture in near real time. The remote sensing–based indicators developed in this study provide a replicable framework for tracking how climate change may alter drought dynamics across regions and over time.

Further extensions to this approach could be considered. This includes .incorporating complementary indicators to distinguish the structural aridity from short-term drought stress and to account for anthropogenic pressures on water resources, including groundwater extraction and land use change. The retrospective analysis of drought exposure could also be strengthened by integrating information on regional economic structures, water infrastructure, and adaptation policies.

Future research could also link remote-sensing drought indicators to economic and social outcomes, such as crop yields, employment, or local government revenues, to better quantify the full costs of drought. Expanding the nowcasting framework to incorporate meteorological and socioeconomic data could further enhance its predictive and policy relevance.

References

Australian Government (2025), *National Water Account: Australia, 2023.*, https://www.bom.gov.au/water/nwa/2023/index.shtml (accessed on 6 October 2025). [30]

Bautista-Capetillo, C. et al. (2016), "Drought Assessment in Zacatecas, Mexico", *Water*, Vol. 8/416, https://doi.org/10.3390/w8100416. [10]

Cárdenas-Tristán, A. et al. (2023), "Spatiotemporal analysis of water reservoirs in San Luis Potosí state, Mexico, from 1990 to 2015", *Water Science*, Vol. 37/1, pp. 344–357, https://doi.org/10.1080/23570008.2023.2259662. [18]

Carrillo-Carrillo, M. et al. (2025), "Spatio-Temporal Analysis of Drought with SPEI in the State of Mexico and Mexico City", *Atmosphere*, Vol. 16/202, https://doi.org/10.3390/atmos16020202. [9]

Conagua (2020), *Programa Nacional Hídrico 2020-2024*, https://www.gob.mx/cms/uploads/attachment/file/642632/PNH_2020-2024__ptimo.pdf (accessed on 8 September 2025). [4]

CONEVAL (2024), *Inventario CONEVAL de programas y acciones federales de desarrollo social 2023*, https://www.coneval.org.mx/Evaluacion/IPFE/Documents/Inventarios_Anteriores/Inventario_2023.zip (accessed on 27 August 2025). [21]

Del-Toro-Guerrero, F. et al. (2022), "Surface Reflectance–Derived Spectral Indices for Drought Detection: Application to the Guadalupe Valley Basin, Baja California, Mexico", *Land*, Vol. 11/783, https://doi.org/10.3390/land11060783. [14]

FAO (2025), *Global Information and Early Warning System: country briefs, Mexico*, https://www.fao.org/giews/countrybrief/country.jsp?code=MEX (accessed on 6 October 2025). [27]

Fuje, H. et al. (2023), "Fiscal Impacts of Climate Disasters in Emerging Markets and Developing Economies", *IMF Working Papers*, Vol. 261, https://doi.org/10.5089/9798400262913.001. [2]

Gu, Y. et al. (2007), "A five-year analysis of MODIS NDVI and NDWI for grassland drought assessment over the central Great Plains of the United States", *Geophysical research letters*, Vol. 34/6. [12]

Méndez, M. and V. Magaña (2010), "Regional Aspects of Prolonged Meteorological Droughts over Mexico and Central America", *Journal of Climate*, Vol. 23/5, pp. 1175–1188, https://doi.org/10.1175/2009JCLI3080.1. [19]

NASA (2025), *Reflected Near-Infrared Waves*, https://science.nasa.gov/ems/08_nearinfraredwaves/ (accessed on 19 September 2025). [17]

OECD (2025), *Climate hazard statistics - Regions (for 'Developer API')*, https://data-explorer.oecd.org/vis?lc=en&df[ds]=dsDisseminateFinalDMZ&df[id]=DSD_REG_CLIM@DF_CLIM&df[ag]=OECD.CFE.EDS (accessed on 11 September 2025). [6]

OECD (2025), *Global Drought Outlook: Trends, Impacts and Policies to Adapt to a Drier World*, https://doi.org/10.1787/d492583a-en. [1]

OECD (2025), *Local Data Portal*, https://localdataportal.oecd.org/maps.html (accessed on 11 September 2025). [5]

OECD (2025), "The Circular Water Economy in Latin America", *OECD Urban Studies, OECD Publishing, Paris*, https://doi.org/10.1787/a0508572-en. [26]

OECD (2024), *10 Years of the OECD Water Governance Initiative*, https://www.oecd.org//content/dam/oecd/en/topics/policy-sub-issues/water-governance/Brochure_10%20years%20of%20the%20OECD%20WGI_PRINT.pdf. [23]

OECD (2024), "A Handbook of What Works: Solutions for the local implementation of the OECD Principles on Water Governance", *OECD Regional Development Papers*, Vol. 72, https://doi.org/10.1787/bf54627e-en. [24]

OECD (2023), "A Territorial Approach to Climate Action and Resilience", *OECD Regional Development Studies, OECD Publishing, Paris*, https://doi.org/10.1787/1ec42b0a-en. [25]

OECD (2018), "Implementing the OECD Principles on Water Governance: Indicator", *OECD Publishing, Paris*, https://doi.org/10.1787/9789264292659-en. [22]

OECD and World Bank (2019), "Fiscal Resilience to Natural Disasters: Lessons from Country Experiences", *OECD Publishing, Paris*, https://doi.org/10.1787/27a4198a-en. [3]

Ortega-Gaucin, D., J. De la Cruz Bartolón and H. Castellano Bahena (2018), "Drought Vulnerability Indices in Mexico", *Water*, Vol. 10/11, https://doi.org/10.3390/w10111671. [20]

Ortiz-Gómez, R. et al. (2018), "Characterization of droughts by comparing three multiscale indices in Zacatecas, Mexico.", *Tecnología y ciencias del agua*, Vol. 9/3, pp. 47-91. [8]

Patil, P. et al. (2024), "Exploration and advancement of NDDI leveraging NDVI and NDWI in Indian semi-arid regions: A remote sensing-based study", *Case Studies in Chemical and Environmental Engineering*, Vol. 9, https://doi.org/10.1016/j.cscee.2023.100573. [13]

Romero, D. et al. (2020), "Standardized Drought Indices for Pre-Summer Drought Assessment in Tropical Areas", *Atmosphere*, Vol. 11/1209, https://doi.org/10.3390/atmos11111209. [7]

Salas-Martínez, F. et al. (2023), "Methodological estimation to quantify drought intensity based on the NDDI index with Landsat 8 multispectral images in the central zone of the Gulf of Mexico", *Frontiers in Earth Science*, Vol. 11/1027483. [15]

USDA (2025), "Grain and Feed Annual: Mexico", https://apps.fas.usda.gov/newgainapi/api/Report/DownloadReportByFileName?fileName=Grain%20and%20Feed%20Annual_Mexico%20City_Mexico_MX2025-0013.pdf. [28]

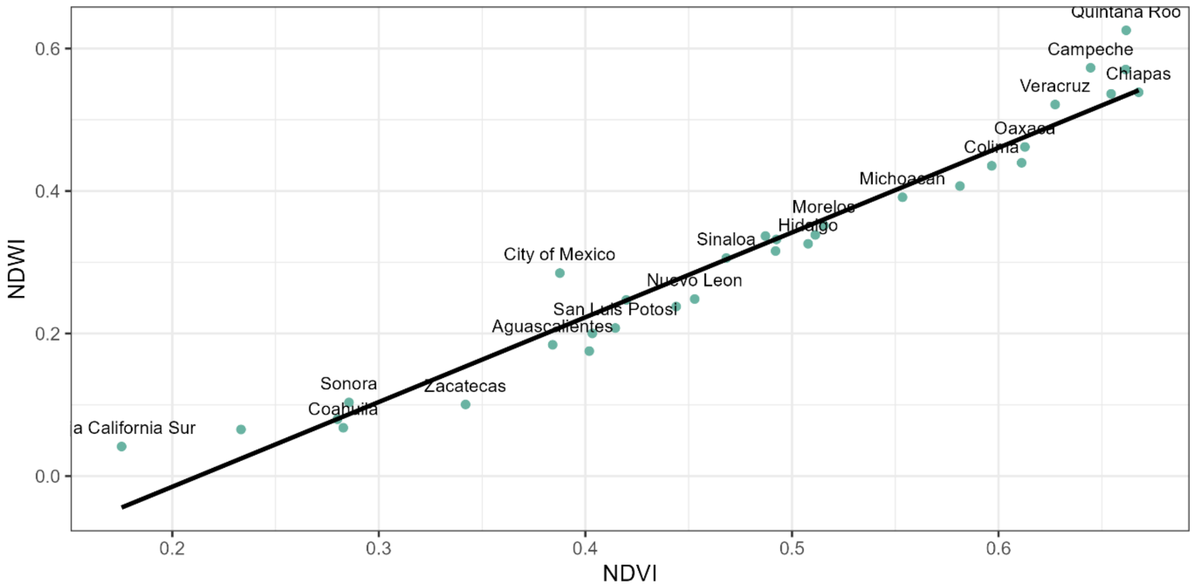
USGS (2025), *MOD09A1.061 Terra Surface Reflectance 8-Day Global 500m*, https://developers.google.com/earth-engine/datasets/catalog/MODIS_061_MOD09A1 (accessed on 8 September 2025). [16]

Velázquez-Zapata, J. and R. Dávila-Ortiz (2025), "Projected Meteorological Drought in Mexico Under CMIP6 Scenarios: Insights into Future Trends and Severity", *Geosciences*, Vol. 15/186, https://doi.org/10.3390/geosciences15050186. [11]

Zahniser, S. et al. (2019), "The growing corn economies of Mexico and the United States", *USDA, the Economic Research Service.*, https://ers.usda.gov/sites/default/files/_laserfiche/outlooks/93633/FDS-19F-01.pdf. [29]

Annex A. High significant correlation between NDVI and NDWI

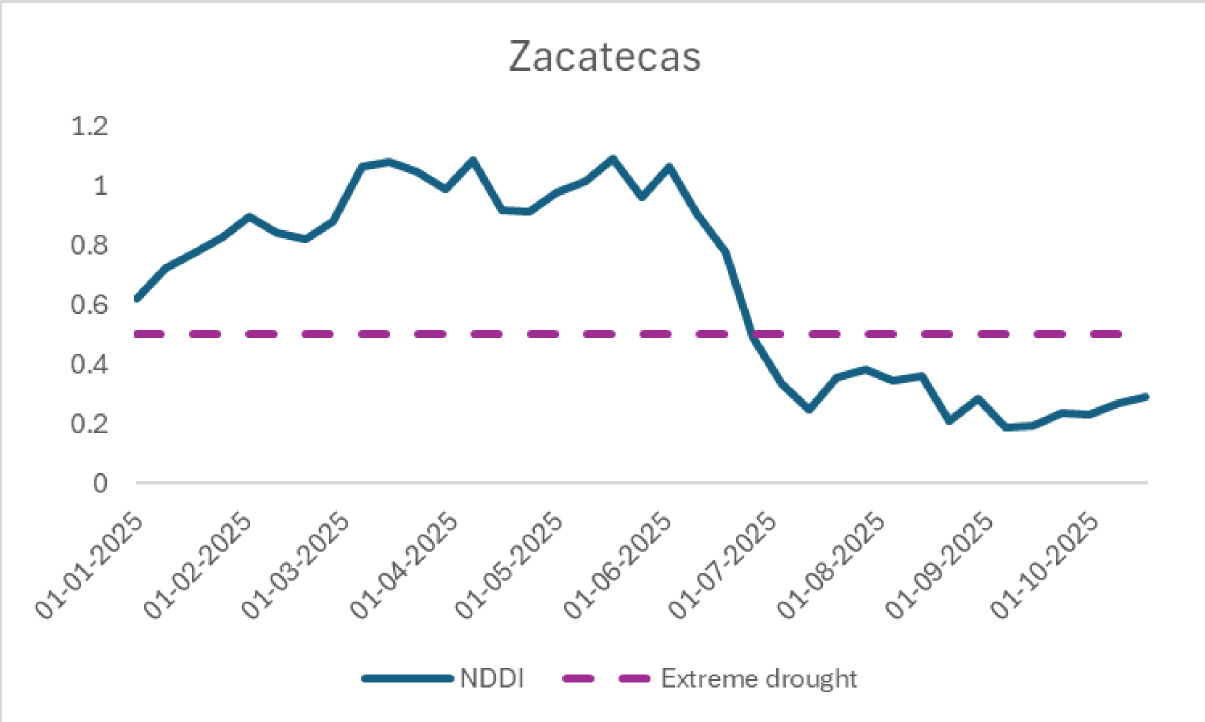
Figure A 1: High significant correlation between NDVI and NDWI across states of Mexico in 2024



Notes: Significant correlation (0.97) at level 0.001 between regional NDVI and NDWI within Mexico.
Source: Authors.

Annex B. The sub-monthly severity of drought does not change overall

Figure A 2: 8 days drought severity in Zacatecas (January 1 to October 24, 2025)



Source: Authors.

Table .1. Main policies implemented in Mexico to reduce drought vulnerability

Fund / Policy / Action	Objectives
Programa Nacional Contra la Sequía (PRONACOSE, National Drought Program)	One of the key policies in Mexico for addressing and managing drought, which integrate technical and administrative preventative measures to minimize environmental, economic, and social impacts. These include the national drought monitoring system, updated every 15 days by the National Meteorological Service (SMN); general agreements regarding the start and end of a state of emergency due to severe weather; the Intersecretariat Commission for Drought and Flood Management (CIASI) with the participation of a total of 14 federal agencies; municipal-level drought vulnerability maps; drought probability maps; and the Drought Prevention and Mitigation Measures Programs (PMPMS) for cities and the Basin Council. Work is underway in 2025 and 2026 to update the technical methodology for drought vulnerability calculation at the municipal level, involving various agencies, including INECC, CENAPRED, CONANP, CONABIO, PROFEPA, IMTA and CONAFOR.
Scientific and technological research	Strength the water management and conservation capacities of the units responsible for implementing water and environmental policy in Mexico through the development of scientific and technological research projects and the provision of postgraduate training through the Mexican Institute of Water Technology.
Research on Climate Change, Sustainability and Green Growth	Provide decision-makers at the institutions comprising the National Climate Change System (SINACC) with up-to-date scientific and technological information and knowledge on climate change, environmental protection, and ecology.
Drinking Water, Drainage and Treatment	Increase the coverage and efficiency of drinking water, sewage, and sanitation services for residents of localities, municipalities, and states in rural and urban areas of the country through financial, technical, and commercial support.
Hydro-agricultural Infrastructure Support Program	Strength infrastructure in agricultural areas, including irrigation districts, irrigation units, and rainfed technical districts through the preservation, rehabilitation, modernization, and expansion of hydro-agricultural infrastructure.
Fondo de Desastres Naturales (FONDEN, Natural Disaster Fund, FONDEN) 1996-2021	Carry out actions, authorize and apply resources to mitigate the effects produced by a disturbing natural phenomenon with two related programs: • Fondo de Prevención de Desastres Naturales (FOPREDEN, Fund for Disaster Prevention). • Fondo de Apoyo Rural por Contingencias Climatológicas (Fund to Support the Rural Population Affected by Climate Hazards).
Plan Nacional Hídrico (PNH, National Water Plan) 2024-2030	Guarantee the human right to water in sufficient quantity and quality, ensure the sustainability of our resources and promote adequate and responsible management of water in all its uses. Districts and irrigation units and the industrial sector commit to the voluntary return of more than 2,500 million cubic meters of water, which they have concessions to be used for human consumption, especially in areas of greatest water stress.

Source: Authors based on (CONEVAL, 2024[21]) and (OECD and World Bank, 2019[3]).